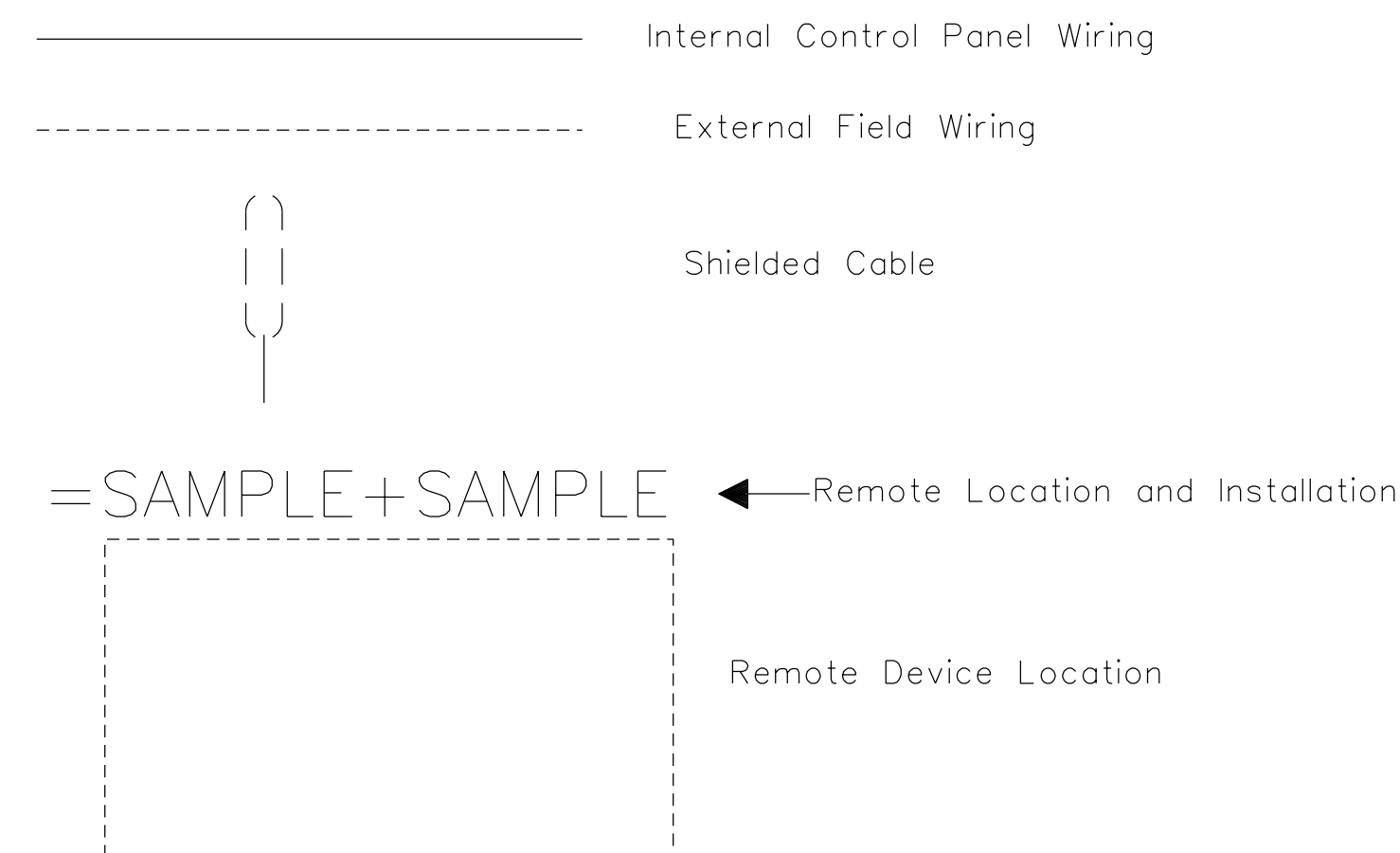
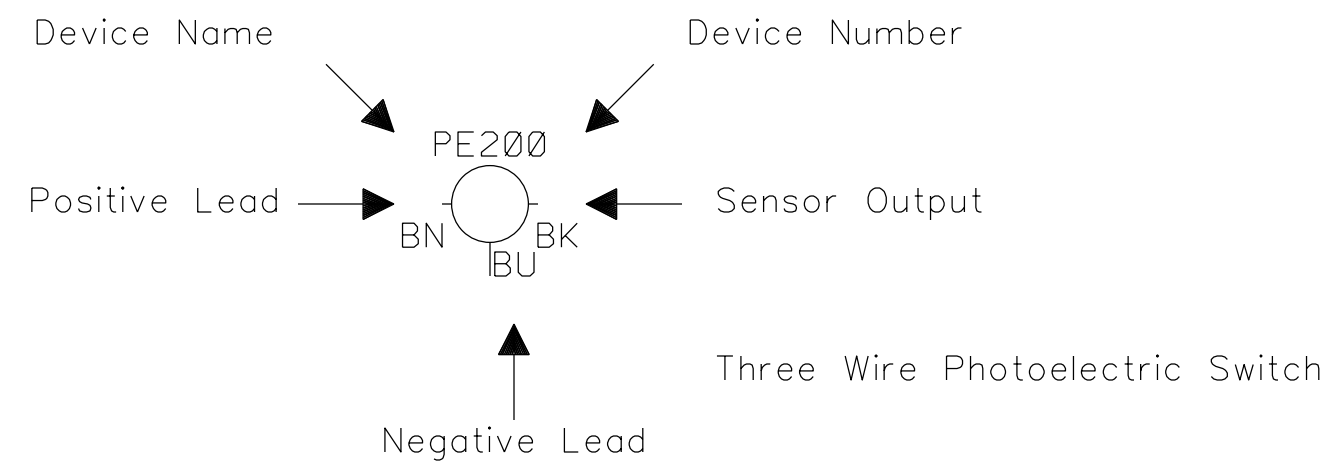
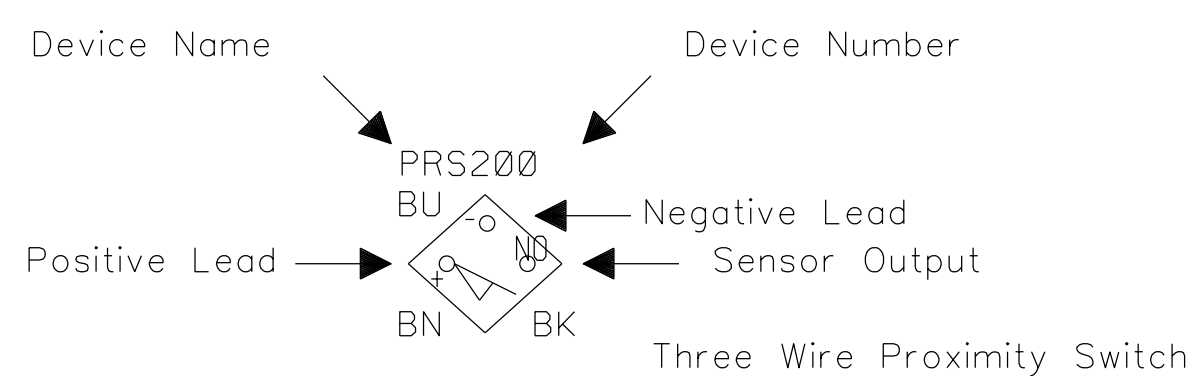
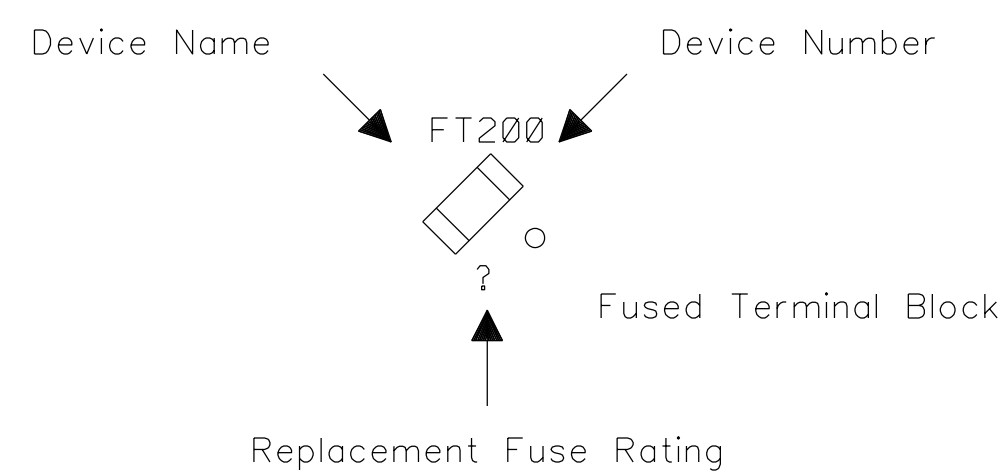
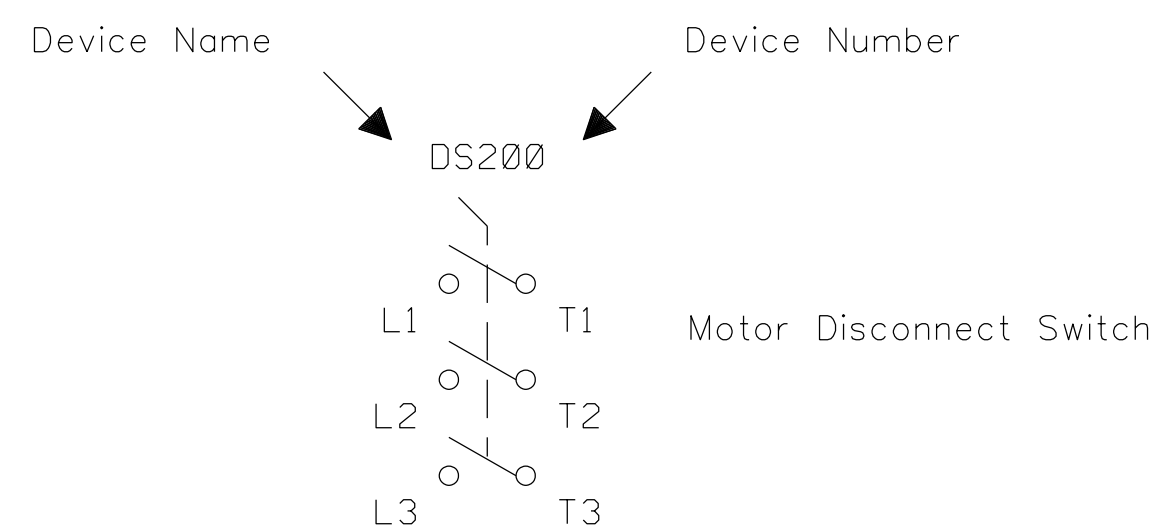


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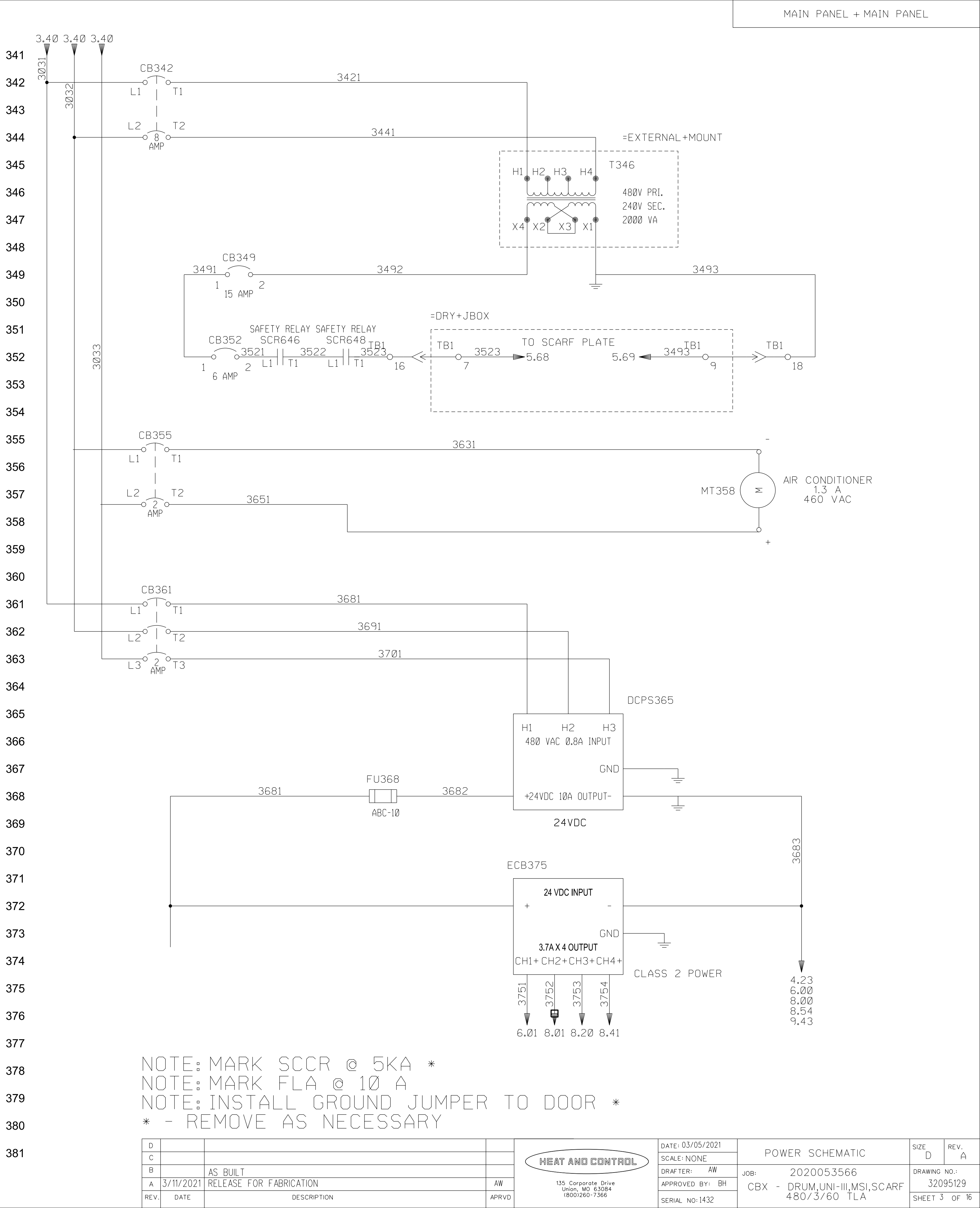
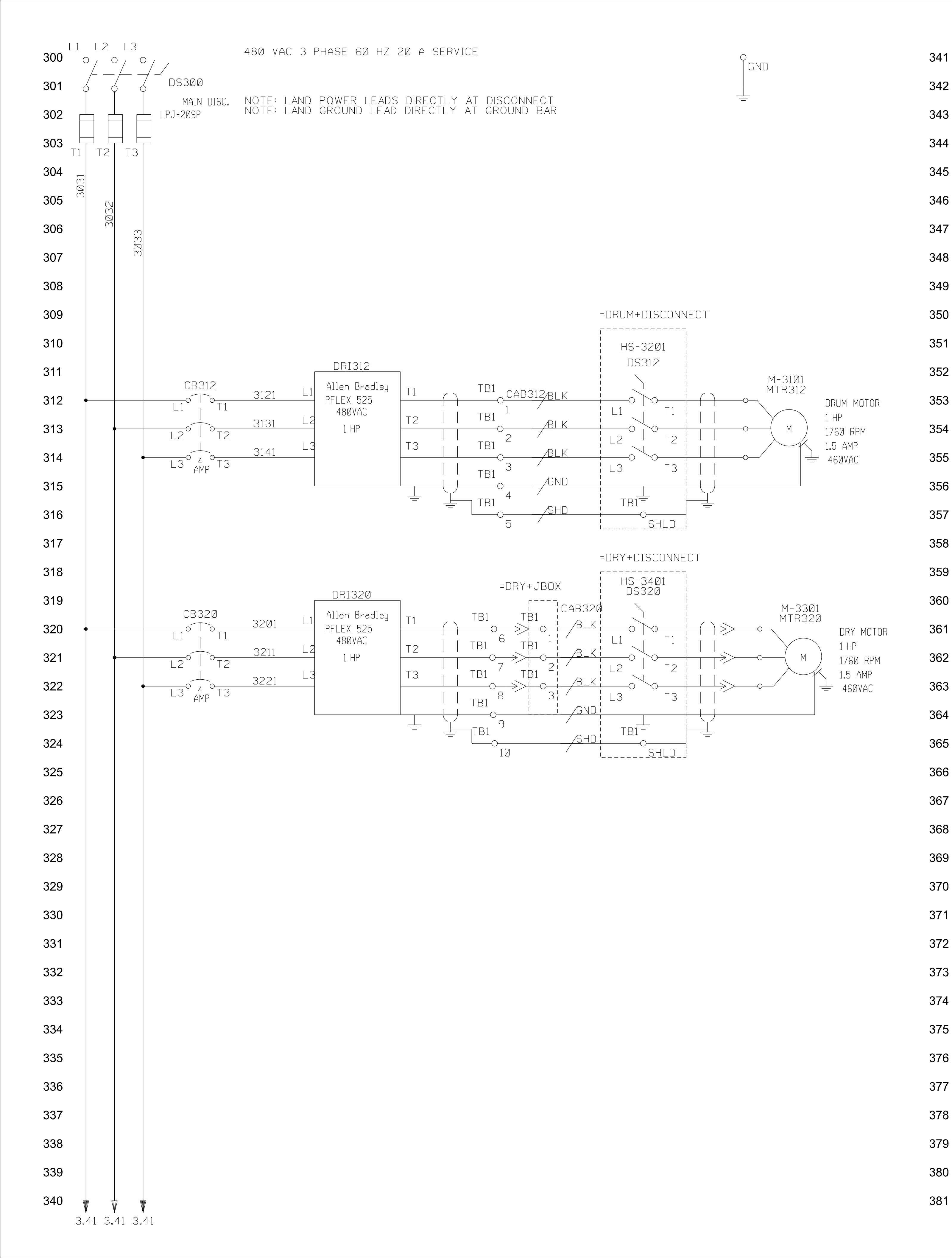
INSTALLATION NAME	DESCRIPTION	NUMBER OF PAGES
INFO	CONTAINS TABLE OF CONTENTS AND DEVICE KEY	1 - 2
CONTROLPANEL	ELECTRICAL SCHEMATICS AND PANEL LAYOUTS	3 - 15
LISTS	BILL OF MATERIAL	16



TO MAINTAIN OVERCURRENT, SHORT-CIRCUIT, AND GROUND-FAULT PROTECTION, THE MANUFACTURER'S INSTRUCTIONS FOR SELECTING CURRENT ELEMENTS AND SETTING THE INSTANTANEOUS-TRIP CIRCUIT BREAKER MUST BE FOLLOWED.

TRIPPING OF THE INSTANTANEOUS-TRIP CIRCUIT BREAKER IS AN INDICATION THAT A FAULT CURRENT HAS BEEN INTERRUPTED. CURRENT-CARRYING COMPONENTS OF THE MAGNETIC MOTOR CONTROLLER SHOULD BE EXAMINED AND REPLACED IF DAMAGED TO REDUCE THE RISK OF FIRE OR ELECTRIC SHOCK. IF BURNOUT OF THE CURRENT ELEMENT OF AN OVERLOAD RELAY OCCURS, THE COMPLETE OVERLOAD RELAY MUST BE REPLACED.

D				HEAT AND CONTROL 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 03/05/21	DEVICE KEY	SIZE D	REV. A
C					SCALE: NONE			
B	AS BUILT				DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129	
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH	CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	SHEET 2 OF 16	
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO:1432			



NOTE: MARK SCCR @ 5KA *

NOTE: MARK FLA @ 10 A

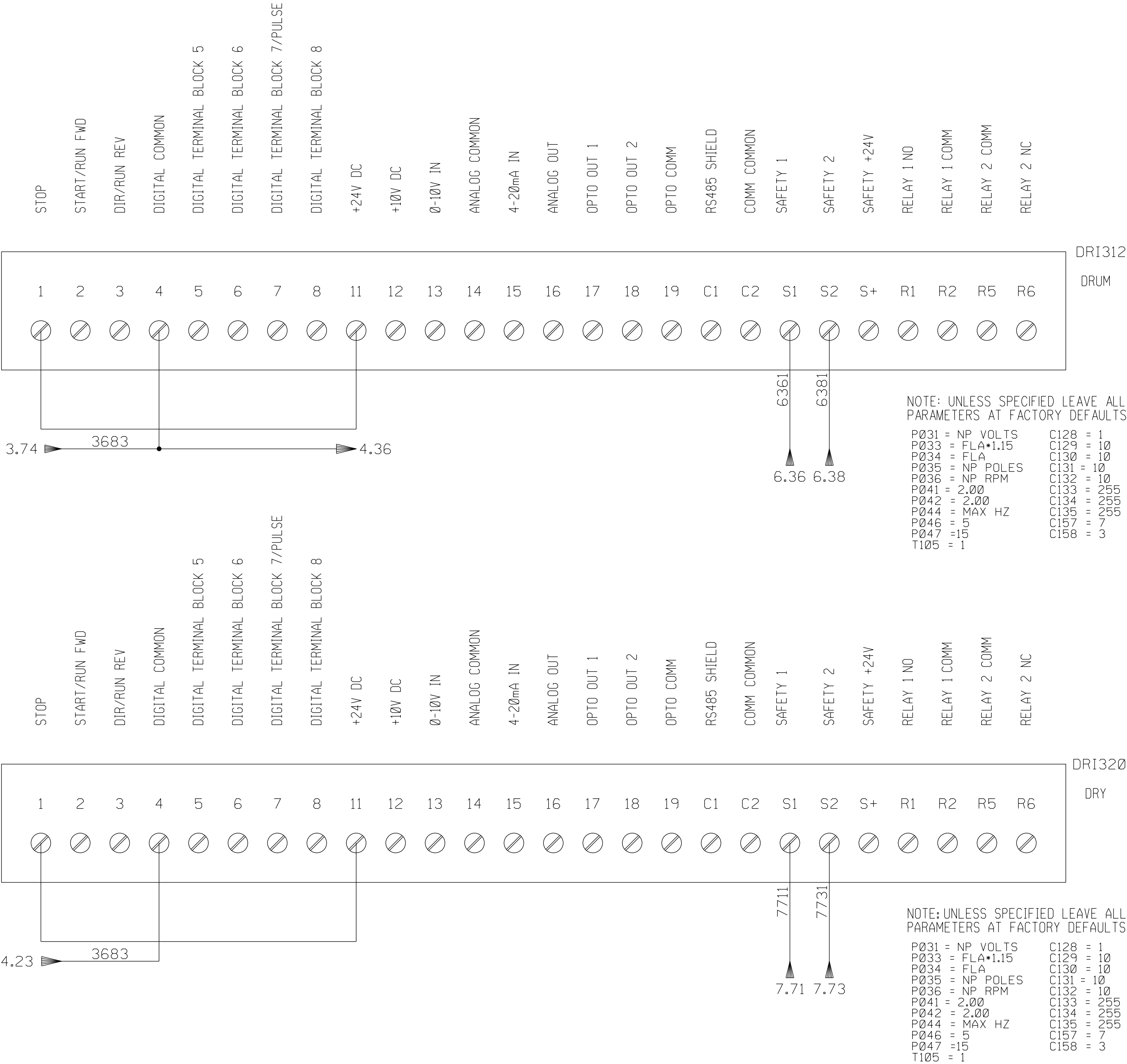
NOTE: INSTALL GROUND JUMPER TO DOOR *

* - REMOVE AS NECESSARY

D				HEAT AND CONTROL 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 03/05/2021	POWER SCHEMATIC		SIZE D	REV. A
C					SCALE: NONE				
B		AS BUILT			DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129		
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH	CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	SHEET 3 OF 16		
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO:1432				

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D				DATE: 03/05/21	DRIVE SCHEMATIC	SIZE	REV.
C				SCALE: NONE		D	A
B		AS BUILT		DRAFTER: AW	2020053566		
A	3/11/2021	RELEASE FOR FABRICATION	AW	APPROVED BY: BH	CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA		DRAWING NO.: 32095129
REV.	DATE	DESCRIPTION	APRVD	135 Corporate Drive Union, MO 63084 (800)260-7366			SHEET 4 OF 16

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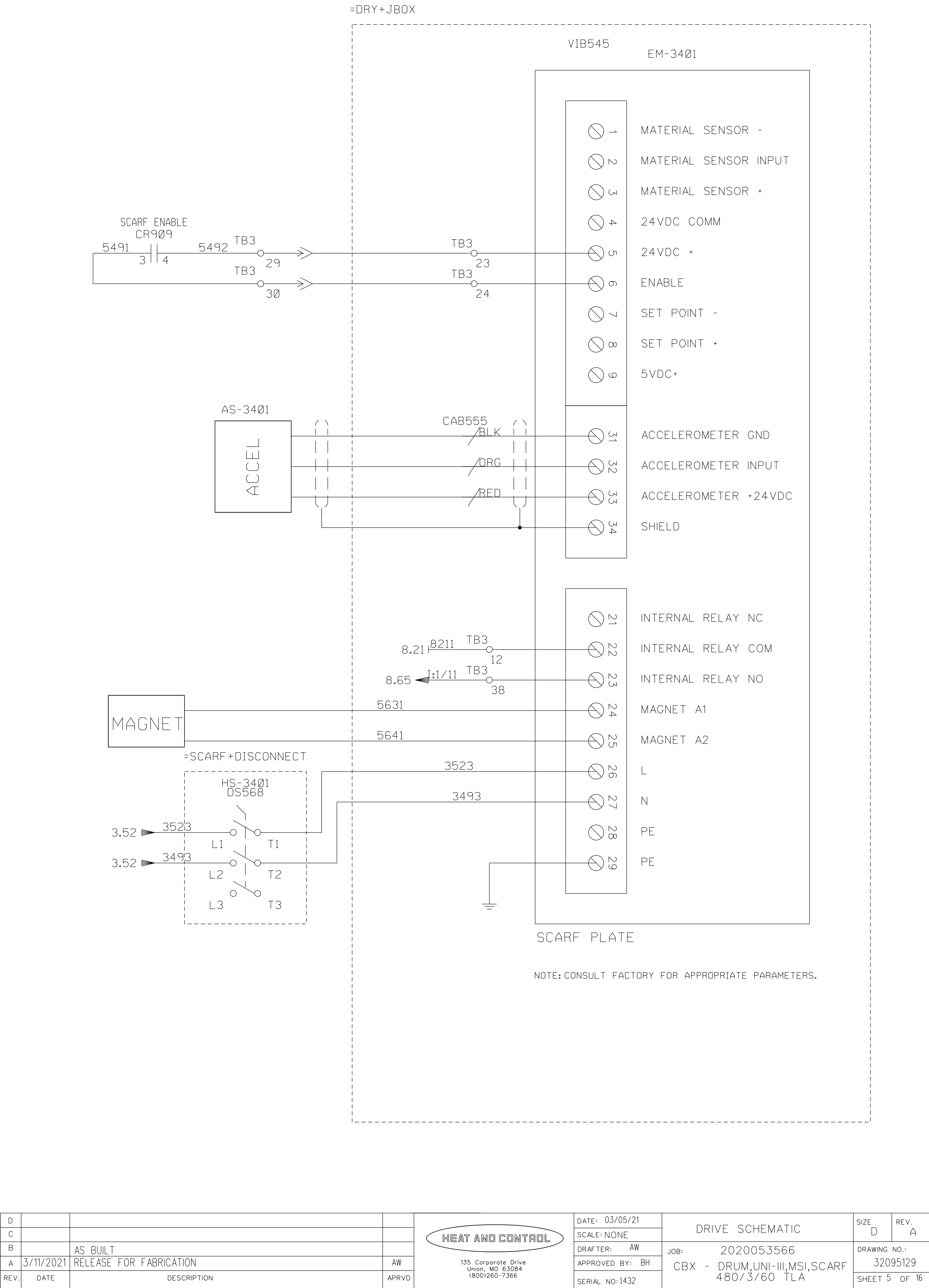
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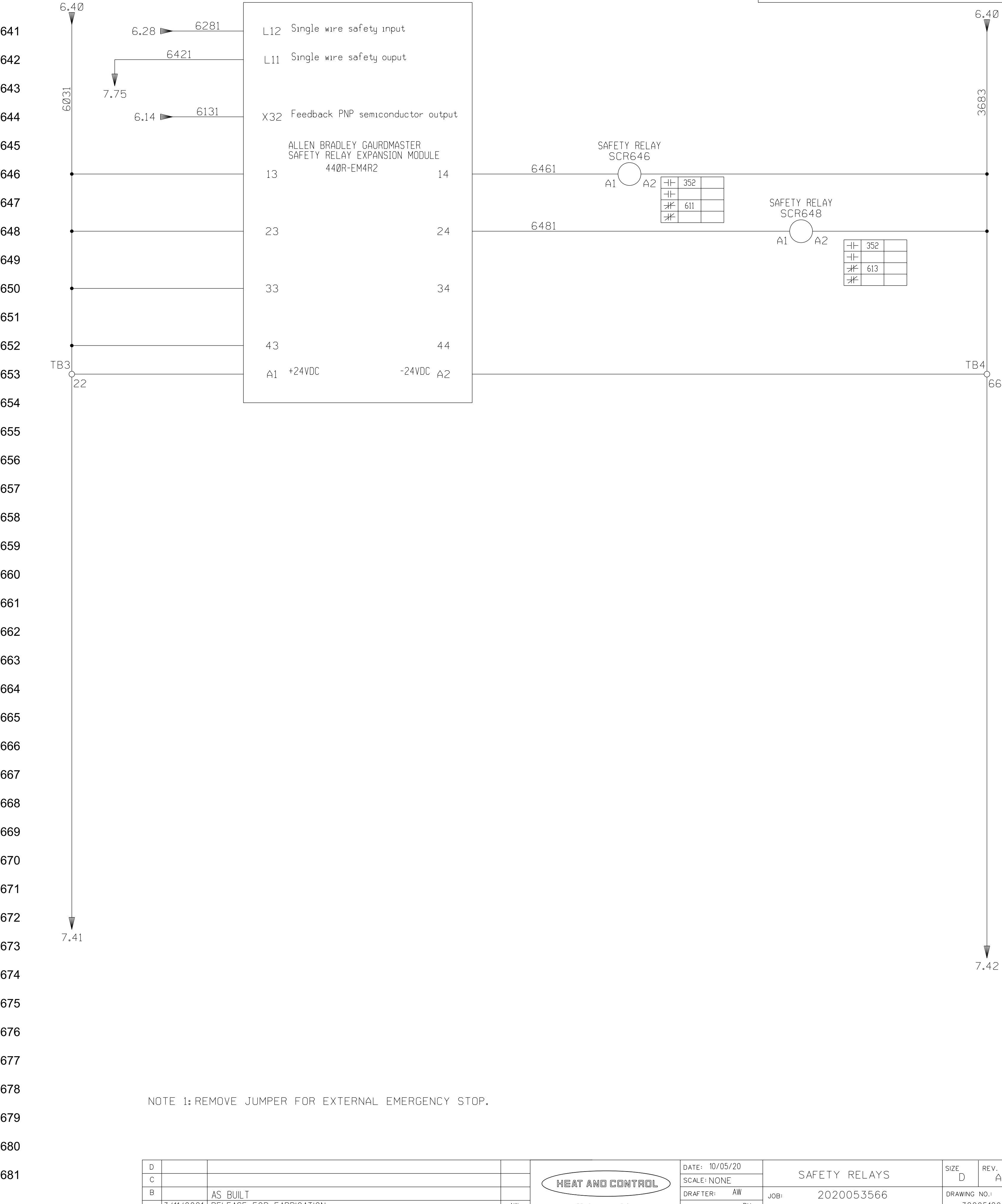
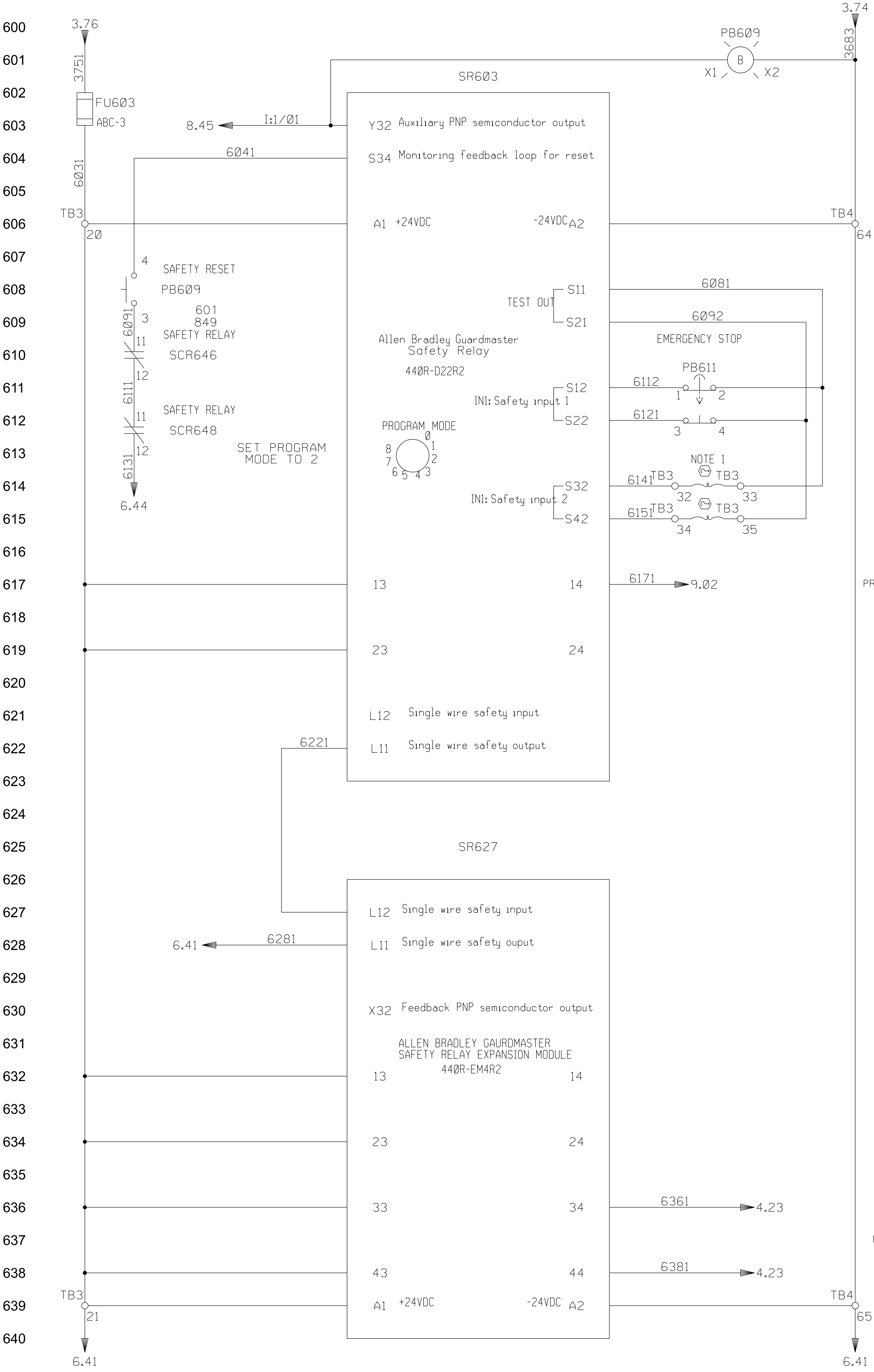
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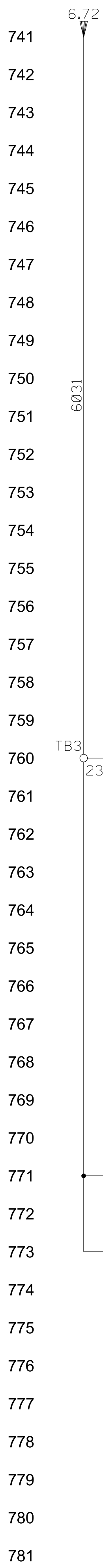
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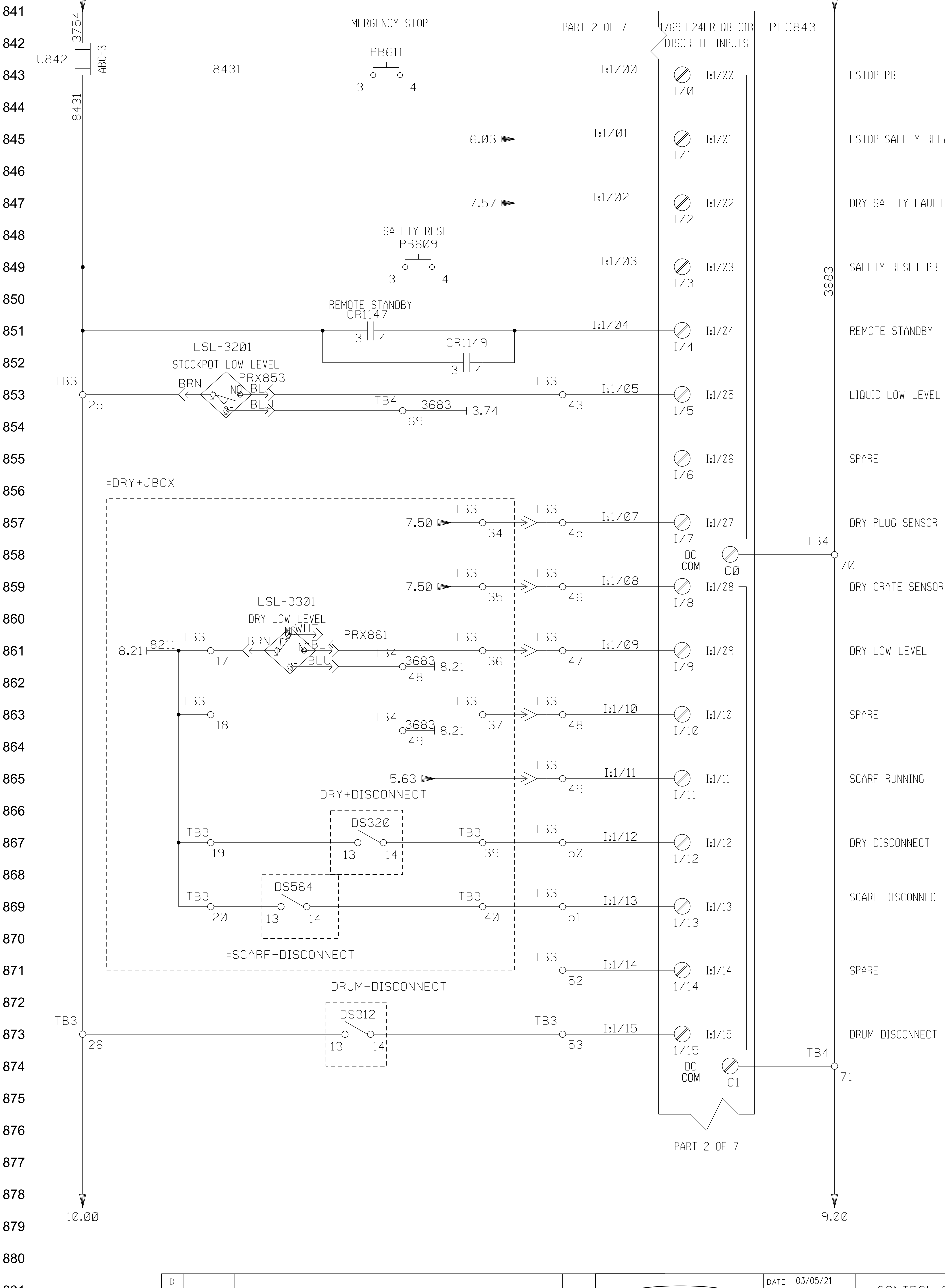
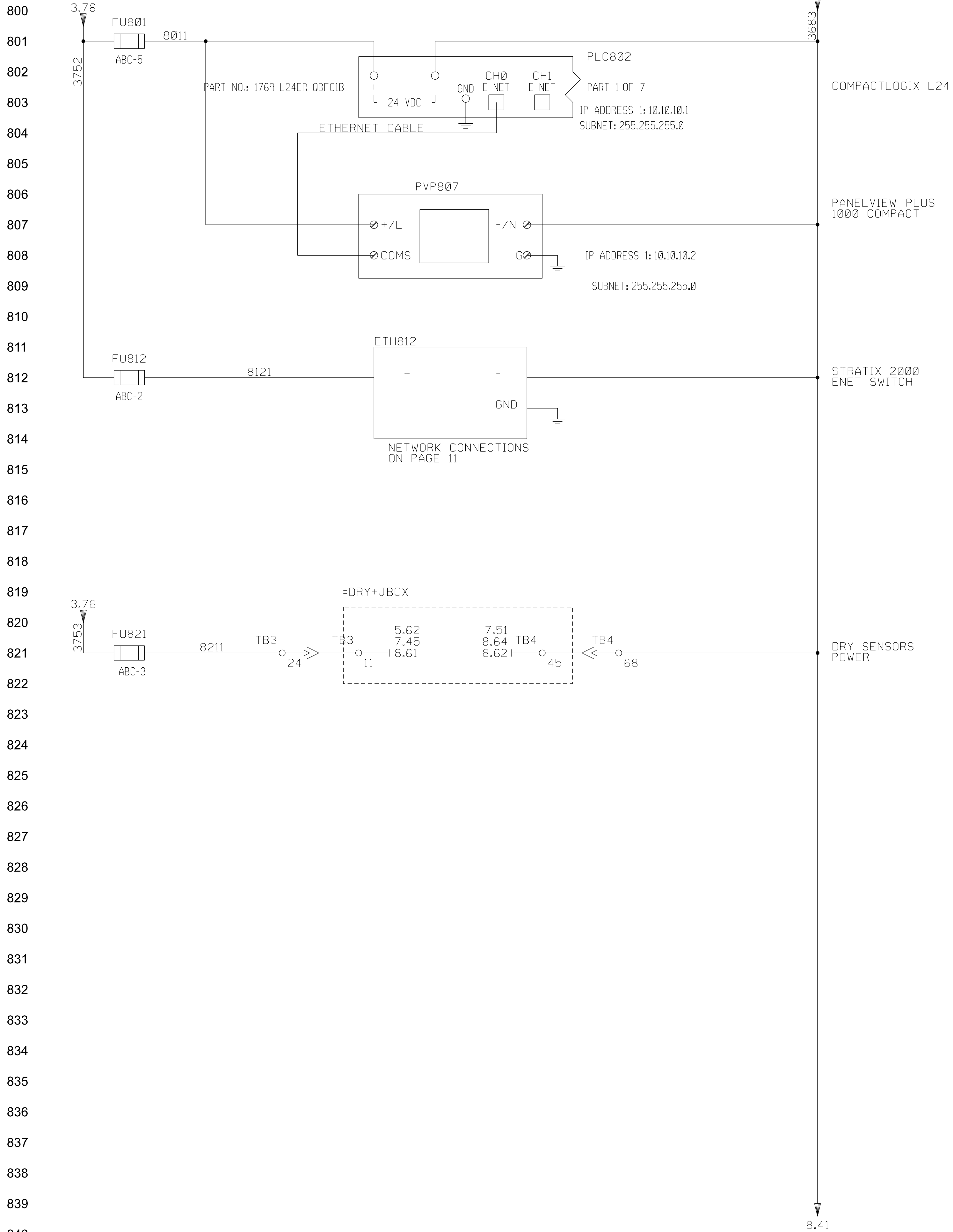
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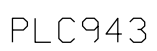




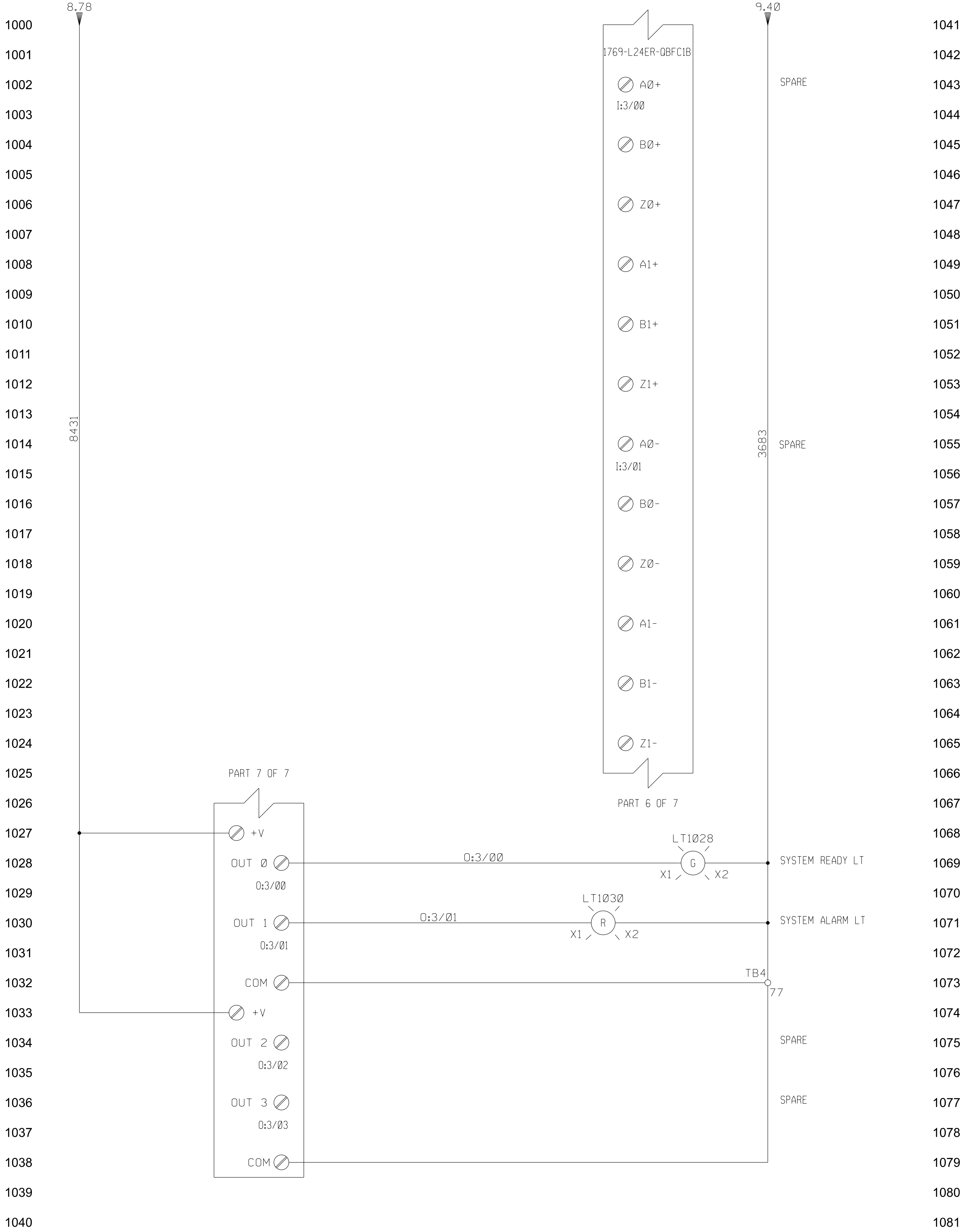
D				HEAT AND CONTROL 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 03/05/21	SAFETY RELAYS		SIZE D	REV. A
C					SCALE: NONE				
B		AS BUILT			DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129		
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH	CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	SHEET 7 OF 16		
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO:1432				



D				HEAT AND CONTROL 155 Corporate Drive Union, MO 65084 (800)260-7366	DATE: 03/05/21	CONTROL SCHEMATICS		SIZE D	REV. A
C					SCALE: NONE				
B		AS BUILT			DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129		
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH	CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	SHEET 8 OF 16		
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO:1432				



D				HEAT AND CONTROL 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 03/05/21	CONTROL SCHEMATICS		SIZE D	REV. A
C					SCALE: NONE				
B		AS BUILT			DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129		
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH			CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO:1432				SHEET 9 OF 16



D				HEAT AND CONTROL 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 10/05/20	CONTROL SCHEMATIC	SIZE	REV.
C					SCALE: NONE		D	A
B		AS BUILT			DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129	
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH			
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO: 1432	CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	SHEET 10 OF 16	

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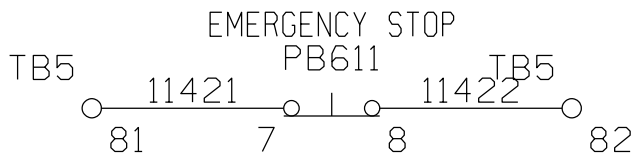
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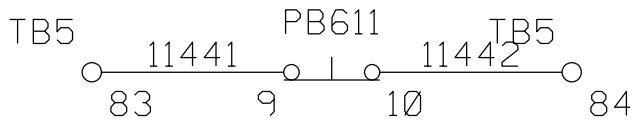
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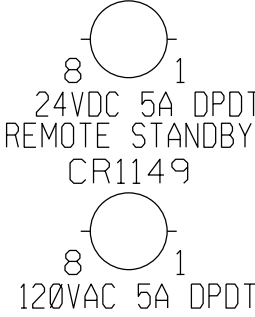
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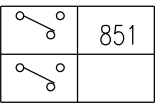
NOTE: ADDITIONAL NC EMERGENCY STOP CONTACTS FOR CUSTOMER USE



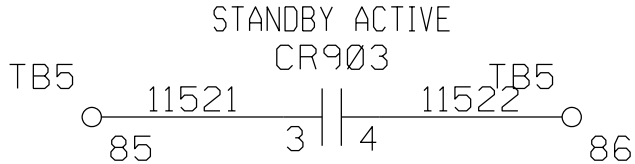
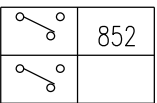
REMOTE STANDBY
CR1147



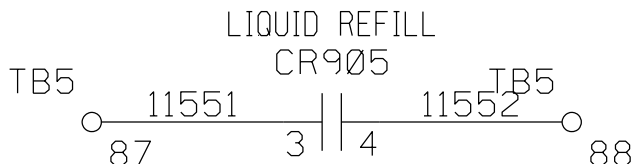
NOTE: WHEN ENERGIZED WITH 24VDC THE SYSTEM WILL ENTER STANDBY



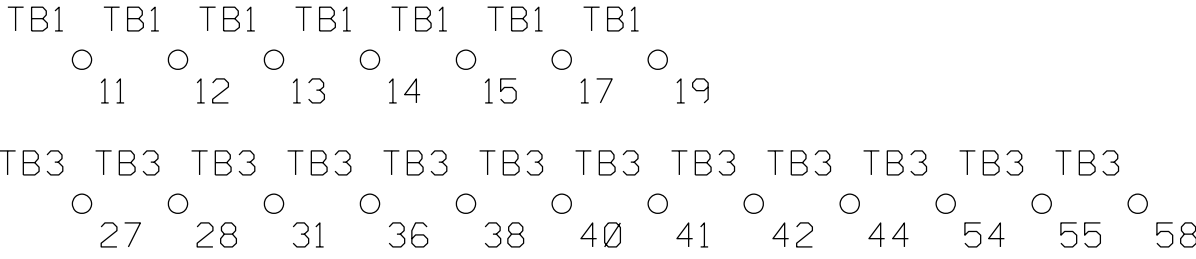
NOTE: WHEN ENERGIZED WITH 120VAC THE SYSTEM WILL ENTER STANDBY



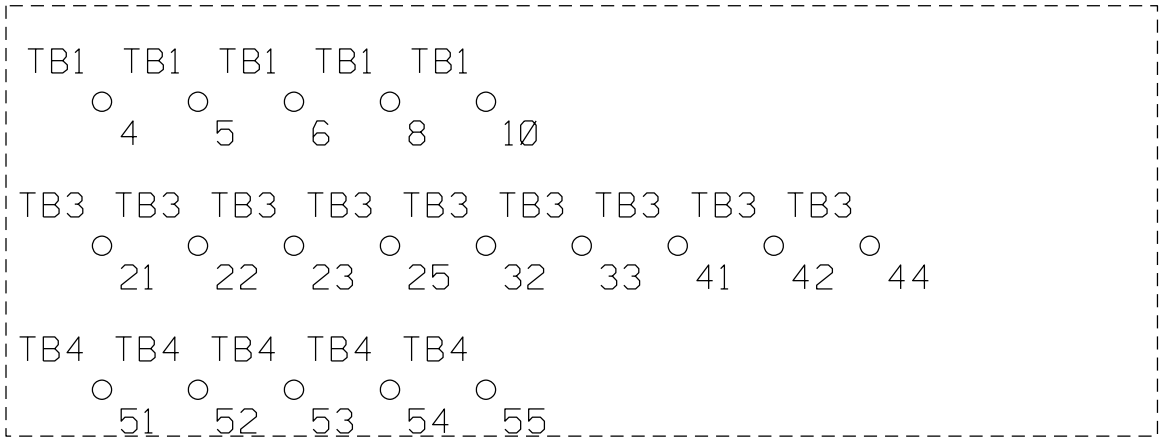
NOTE: CONTACTS RATED AT 5A MAX



SPARE TERMINALS



=DRY+JBOX

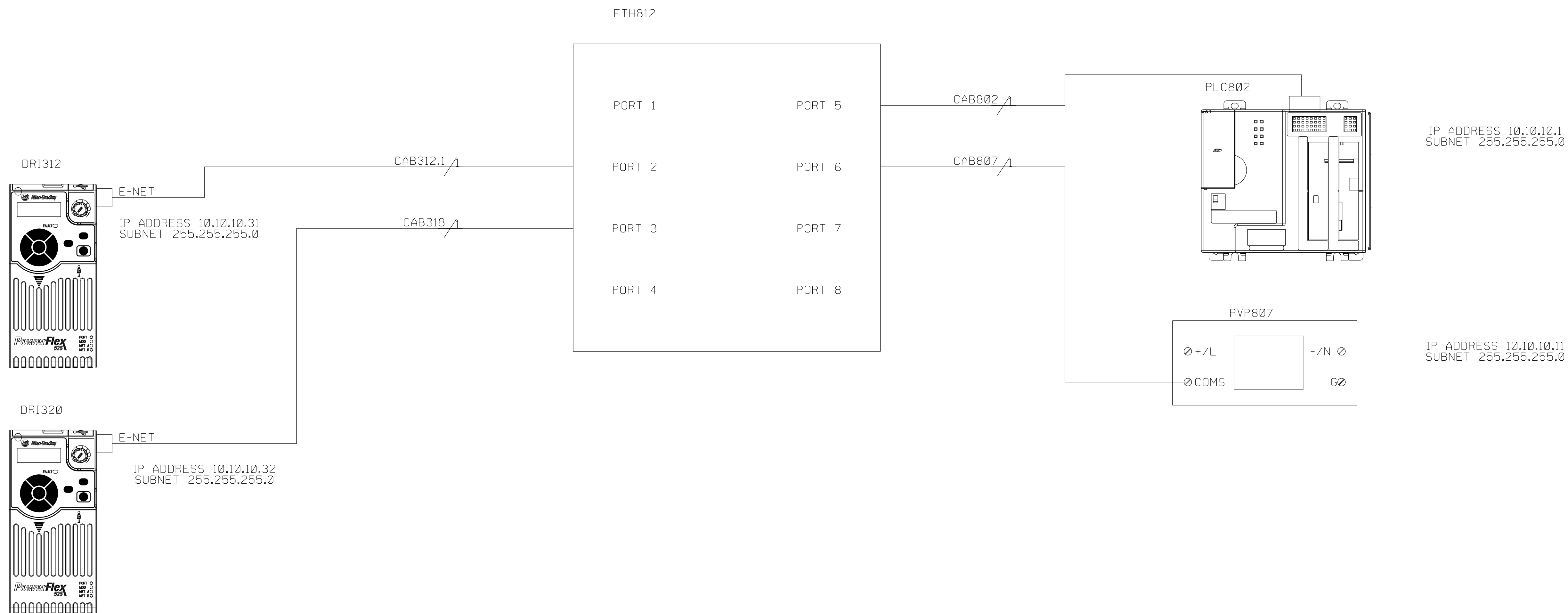


D			
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B		AS BUILT	
A	3/11/2021	RELEASE FOR FABRICATION	AW
REV.	DATE	DESCRIPTION	APRVD

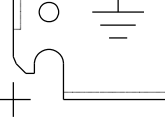
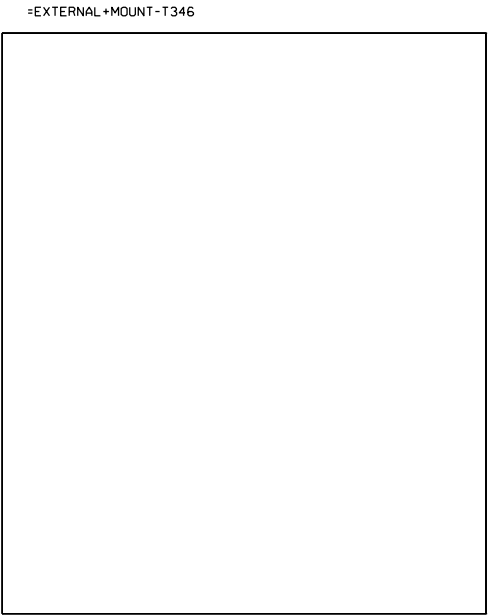
HEAT AND CONTROL

135 Corporate Drive
Union, MO 63084
(800)260-7366

DATE: 03/05/21	CONTROL SCHEMATIC	SIZE D	REV. A
SCALE: NONE			
DRAFTER: AW			
APPROVED BY: BH			
SERIAL NO: 1432	JOB: 2020053566 CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	DRAWING NO.: 32095129	SHEET 11 OF 16

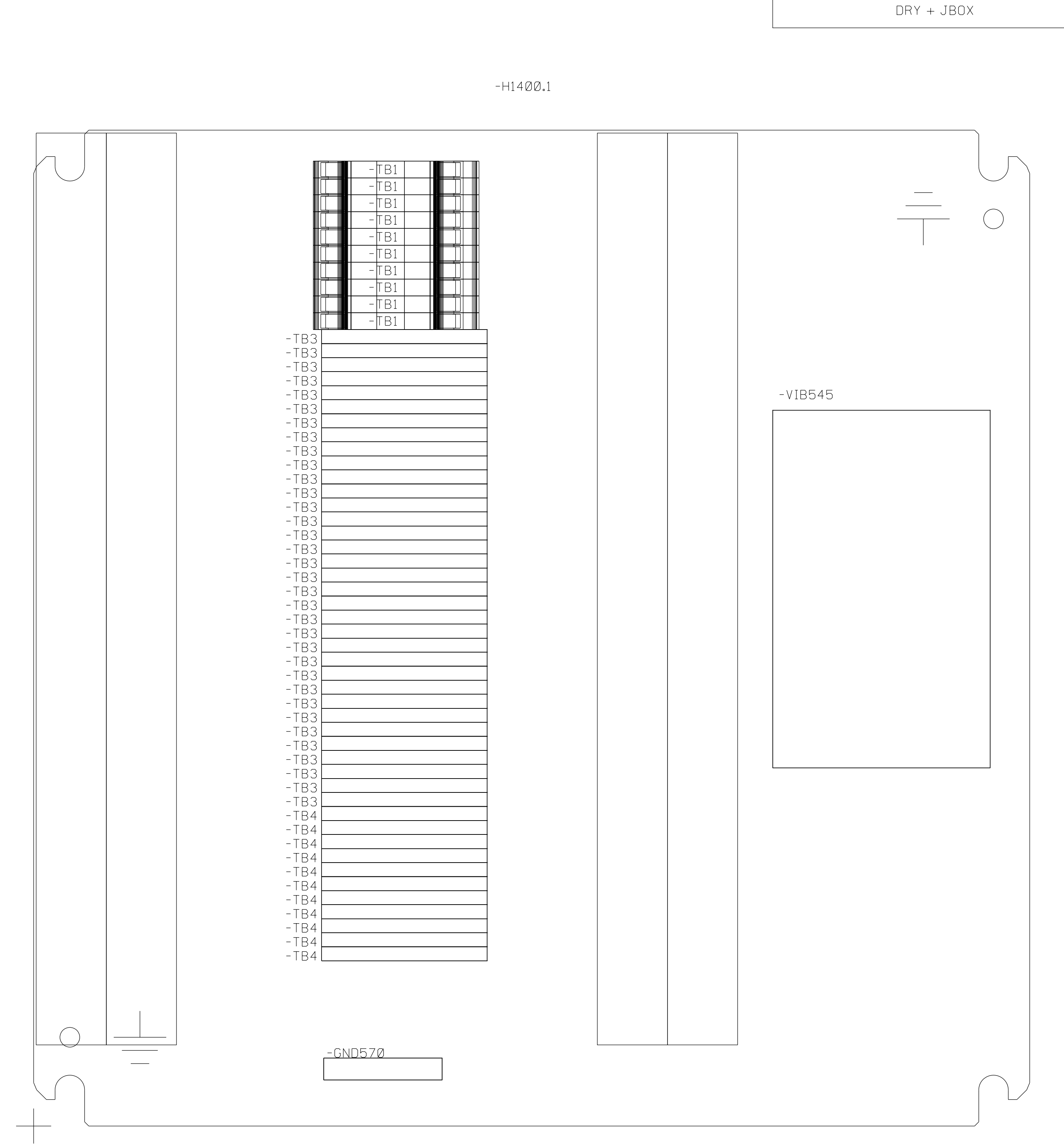
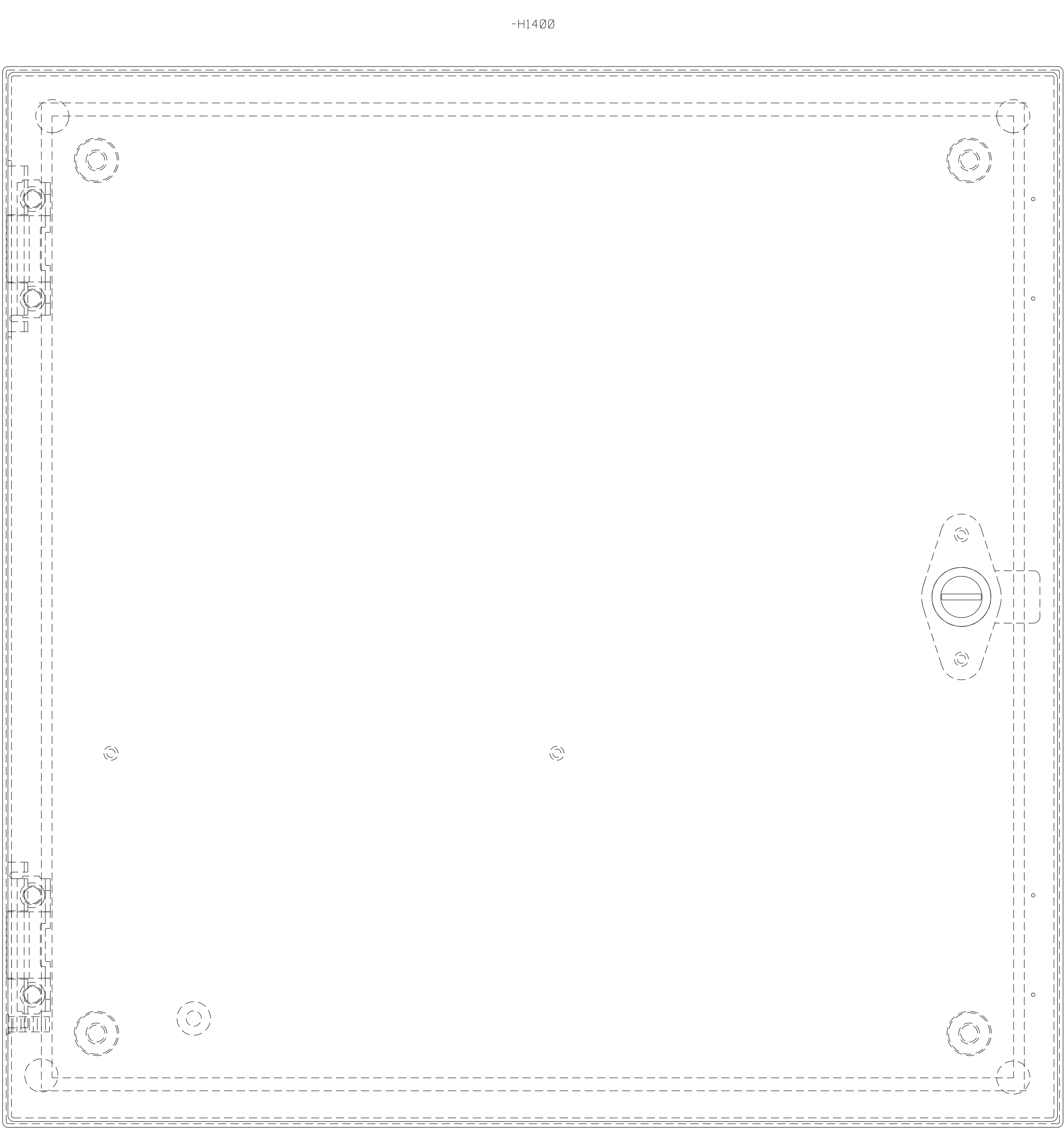


D				 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 03/05/21	ETHERNET NETWORK LAYOUT	SIZE	REV.
C					SCALE: NONE		D	A
B		AS BUILT			DRAFTER: AW		JOB: 2020053566	DRAWING NO.:
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH		CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	32095129
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO:1432			SHEET 12 OF 16



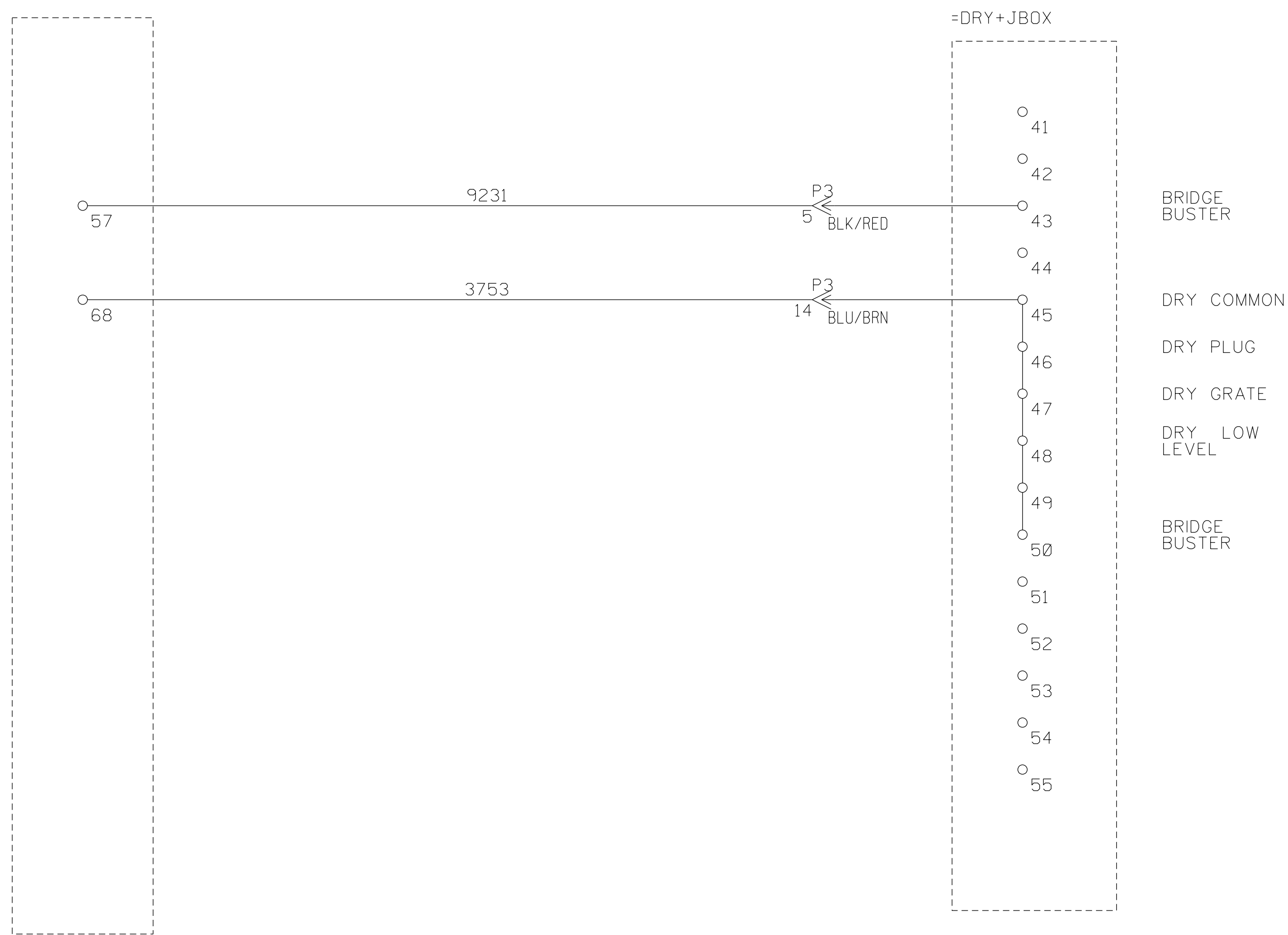
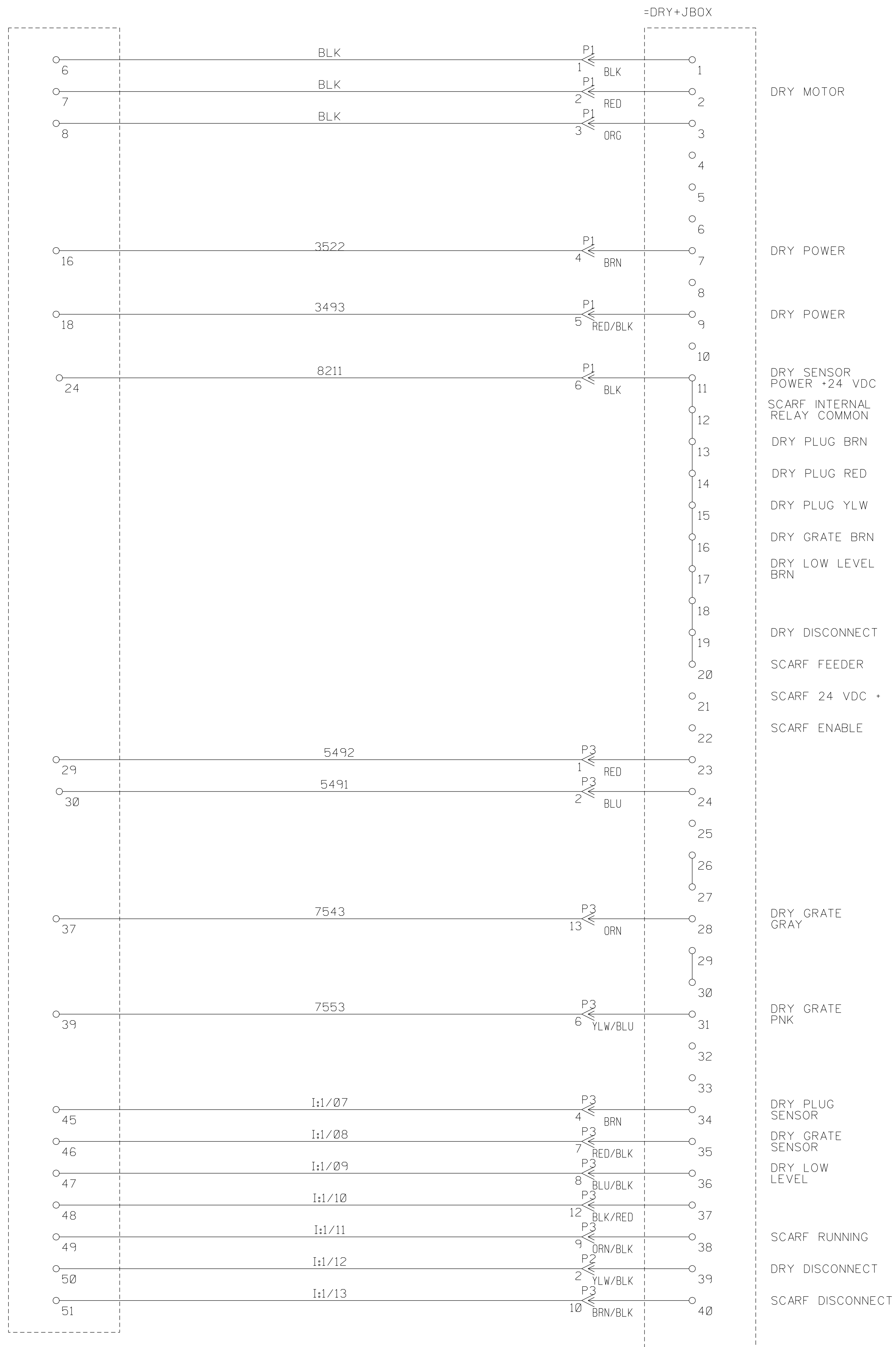
NOTE: REMOVE ALL PAINT FROM BEHIND GROUND BAR.
NOTE: 95 TOTAL TERMINALS IN PANEL
NOTE: LEAVE FIRST ETHERNET PORT OPEN ON SWITCH AND PLC

D				HEAT AND CONTROL 155 Corporate Drive Union, MO 65084 (800)260-7366	DATE: 03/05/21	MAIN PANEL LAYOUT	SIZE	REV.
C					SCALE: NONE		D	A
B	AS BUILT				DRAFTER: AW			
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH			
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO:1432			
						JOB: 2020053566 CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	DRAWING NO.: 32095129 SHEET 13 OF 16	



NOTE: REMOVE ALL PAINT FROM BEHIND GROUND BAR.
NOTE: 55 TOTAL TERMINALS IN PANEL

D				HEAT AND CONTROL 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 03/05/21	DRY PANEL LAYOUT	SIZE	REV.
C					SCALE: NONE		D	A
B		AS BUILT			DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129	
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH			
REV.	DATE	DESCRIPTION	APRVD		SERIAL NO: 1432	CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA	SHEET 14 OF 16	



D				HEAT AND CONTROL 135 Corporate Drive Union, MO 63084 (800)260-7366	DATE: 03/05/21	INTERCONNECT	SIZE D	REV. A
C					SCALE: NONE			
B	AS BUILT				DRAFTER: AW	JOB: 2020053566	DRAWING NO.: 32095129	
A	3/11/2021	RELEASE FOR FABRICATION	AW		APPROVED BY: BH			CBX - DRUM,UNI-III,MSI,SCARF 480/3/60 TLA
REV.	DATE	DESCRIPTION	APRVD	SERIAL NO: 1432		SHEET 15 OF 16		

Device Tag	Qty.	Part Number	Manufacturer	MFG Part Number	Description
CAB312.1	1	20574542	ALLEN BRADLEY	1585J-M8HBJM-2	E-NET
CAB318	1	20574542	ALLEN BRADLEY	1585J-M8HBJM-2	E-NET
CAB802	1	20574542	ALLEN BRADLEY	1585J-M8HBJM-2	E-NET
CAB807	1	20574542	ALLEN BRADLEY	1585J-M8HBJM-2	E-NET
CB312	1	20526627	ALLEN-BRADLEY	140M-C2E-B40	CIRCUIT BREAK, MOTOR PROT,A-B#140MC2EB40
CB320	1	20526627	ALLEN-BRADLEY	140M-C2E-B40	CIRCUIT BREAK, MOTOR PROT,A-B#140MC2EB40
CB342	1	20595076	ALLEN BRADLEY	1489-M2C-080	circuit breaker, 8A, double pole, trip class C,480V
CB349	1	20595108	ALLEN BRADLEY	1489-M1C150	CIRCUIT BREAK, 15A,1P,480V
CB352	1	20595105	ALLEN BRADLEY	1489-M1C060	CIRCUIT BREAK, 6A,1P,480V
CB355	1	20598030	ALLEN BRADLEY	1489-M2D020	circuit breaker, 2A, double pole, trip class D,480V
CB358	1	20595105	ALLEN BRADLEY	1489-M1C060	CIRCUIT BREAK, 6A,1P,480V
CB361	1	20598036	ALLEN BRADLEY	1489-M3D020	CIRCUIT BREAKER, 2A, 3 POLE, 480V
CR903	1	20561979	ALLEN BRADLEY	700-HN122	RELAY SOCKET
CR903	1	20547101	ALLEN BRADLEY	700-HK32Z24	AB HK DPDT 24V
CR905	1	20561979	ALLEN BRADLEY	700-HN122	RELAY SOCKET
CR905	1	20547101	ALLEN BRADLEY	700-HK32Z24	AB HK DPDT 24V
CR907	1	20561979	ALLEN BRADLEY	700-HN122	RELAY SOCKET
CR907	1	20547101	ALLEN BRADLEY	700-HK32Z24	AB HK DPDT 24V
CR909	1	20561979	ALLEN BRADLEY	700-HN122	RELAY SOCKET
CR909	1	20547101	ALLEN BRADLEY	700-HK32Z24	AB HK DPDT 24V
CR1147	1	20561979	ALLEN BRADLEY	700-HN122	RELAY SOCKET
CR1147	1	20547101	ALLEN BRADLEY	700-HK32Z24	AB HK DPDT 24V
CR1149	1	20561979	ALLEN BRADLEY	700-HN122	RELAY SOCKET
CR1149	1	20547103	ALLEN BRADLEY	700-HK32A1	120V 5A DPDT 700HK
DCPS365	1	20571727	SOLA HEVI DUTY	SDN-10-24-480	3PHASE 24V SUPPLY 10A
DRI312	1	20581229	ALLEN BRADLEY	25B-D2P3N104	FREQ DRIVE 1HP 480V 3PH
DRI320	1	20581229	ALLEN BRADLEY	25B-D2P3N104	FREQ DRIVE 1HP 480V 3PH
DS300	1	20562014	ALLEN BRADLEY	194R-J30-1753-PYS1	30A FUSED DISC KIT
DS300	3	20562504	BUSSMANN	LPJ-20SP	LPJ-20SP
ECB375	1	20632877	ALLEN BRADLEY	AB 1692-ZRCLSS	PROTECTOR
ETH812	1	20558930	ALLEN-BRADLEY	1783-US8T	8 PORT ETHERNET SWITCH
FU368	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU368	1	20563681	BUSSMANN	ABC-10	ABC-10
FU603	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU603	1	20562470	BUSSMANN	ABC-3	ABC-3
FU801	1	20562016	ALLEN BRADLEY	1492-N37	END BARRIER FOR AGC
FU801	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU801	1	20562487	BUSSMAN	ABC-5	ABC-5
FU812	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU812	1	20547107	BUSSMANN	ABC-2	ABC-2
FU821	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU821	1	20562470	BUSSMANN	ABC-3	ABC-3
FU842	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU842	1	20562470	BUSSMANN	ABC-3	ABC-3
FU921	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU921	1	20547106	BUSSMANN	ABC-1	ABC-1
FU923	1	20562026	ALLEN BRADLEY	1429-H5	AGC FUSE BLOCK W/24VDC BLOWN IND
FU923	1	20547106	BUSSMANN	ABC-1	ABC-1
GND570	1	20574707	SQUARE D	PK27GTACU	GROUND,EQ,BAR,27 TERM
H1300	1	20565779	HOFFMAN	CSD423612SS	CONCEPT 42X36X12
H1300	1	20562760	HOFFMANN	C-MFKSS	MOUNTING FEET
H1300.1	1	20567259	HOFFMANN	C-P4236	CONCEPT 42X36
H1400	1	20562396	HOFFMAN	CSD161610SS	CONCEPT SS 16X16X10
H1400	1	20562760	HOFFMANN	C-MFKSS	MOUNTING FEET
H1400.1	1	20562934	HOFFMANN	C-P1616	CONCEPT 16X16
LT1028	1	20562730	ALLEN BRADLEY	800H-QR24G	GREEN 4X 24V LIGHT
LT1028	2	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
LT1030	1	20562726	ALLEN BRADLEY	800H-QR24R	RED 4X 24V LIGHT
LT1030	2	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
MT358	1	20597949	HOFFMAN	CR230226G037	AIR CONDITIONER,ENCLOSURE
PB609	2	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
PB609	1	20633126	ALLEN BRADLEY	800H-QRBH2W	SWITCH,WHITE PB ILLU
PB609	1	20562007	ALLEN BRADLEY	800T-XD3	N.O. CONTACT AB 800T
PB609	2	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
PB611	1	20582362	ALLEN BRADLEY	800H-FRXT6A5	SWITCH, PSH/PLL BUT, RED
PB611	4	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
PB611	1	20563017	ALLEN BRADLEY	800F-15YE112	PLATE, ESTOP LEGEND FOR CE
PB611	2	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
PB611	1	20508474	ALLEN BRADLEY	800TC-XA	N.C./N.O. AB 800T
PB611	1	20562008	ALLEN BRADLEY	800T-XD4	N.C. CONTACT AB 800T
PB611	2	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
PB758	2	20562314	ALLEN BRADLEY	800T-N322	SCREW COVER
PB758	1	20633126	ALLEN BRADLEY	800H-QRBH2W	SWITCH,WHITE PB ILLU
PLC802	1	20578470	ALLEN BRADLEY	1769-L24ER-QBFC1B	CPACTLOGIX PROCESSOR
PVP807	1	20584905	EATON	SD-A1-S	SD CARD
PVP807	1	20596454	ALLEN BRADLEY	2711P-T10C22D9P	PV PLUS 1000 COLOR/TOUCH
SCR646	1	20573784	Allen Bradley	100S-C16DJ404C	SAFETY CONTACTOR,4POLE,24VDC COIL,16AMP

Device Tag	Qty.	Part Number	Manufacturer	MFG Part Number	Description
SCR648	1	20573784	Allen Bradley	100S-C16DJ404C	SAFETY CONTACTOR,4POLE,24VDC COIL,16AMP
SR603	1	20581216	ALLEN BRADLEY	440R-D22R2	SAFETY RELAY
SR627	1	20581219	ALLEN BRADLEY	440R-EM4R2	SAFETY RELAY EXPANSION
SR641	1	20581219	ALLEN BRADLEY	440R-EM4R2	SAFETY RELAY EXPANSION
SR757	1	20581216	ALLEN BRADLEY	440R-D22R2	SAFETY RELAY
T346	1	20605944	SOLA HEVI DUTY	HSS14F2BS4X	TRANSFORMER 2KVA 4X SS
VIB545	1	20562349	REO-USA INC	621614	UL VIB FEEDER CONTROLLER, MFS268 6AMP, IP20